

Number of Questions: 4

Answers of Midterm Exam

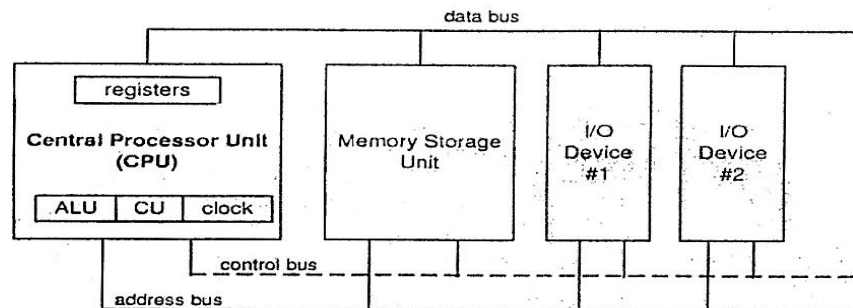
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Name:.....

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Answer the following questions:

- 1- Draw a block diagram of a microcomputer, and then describe the functions of the registers IR and MAR? (4 marks)

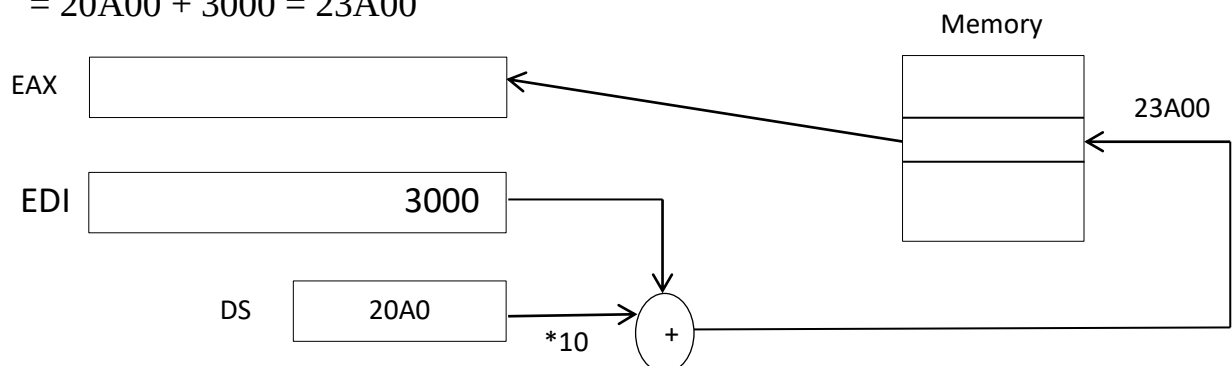


IR (Instruction Register): Contains the instruction most recently fetched.

MAR (Memory Address Register): Contains a word of data to be written to memory or the word most recently read.

- 2- Given that SS = 10C0H, DS = 20A0H, DI = 3000H and BP = 0100H. Calculate and show by figure the memory address accessed by the following instruction: MOV EAX, [DI] in direct addressing environment. (6 marks)

$$\begin{aligned} \text{Address} &= (\text{DS} \times 10) + \text{DI} \\ &= 20\text{A}00 + 3000 = 23\text{A}00 \end{aligned}$$



1- An assembler is

- a) easy to run and good with small programs b) difficult to run but good with small programs c) easy to run and good with long programs d) difficult to run and good with long programs

2- General purpose registers are

- a) 16 bit registers such as CS, DS b) 32 bit registers such as EFLAGS c) 32 bit registers such as ECX, EDX d) None

3- The operands in MOV instruction, source and destination in an instruction cannot be

- a) register, register b) memory location, memory location c) memory location, register d) immediate data, register

4- The segment register of the instruction MOV BX, 50[BP] will be:

- a) SS b) CS c) DS which contains 50 d) ES

5- PROM stands for:

- a) programmable read only memory b) programmable read write memory c) portable read only memory d) None of the mentioned

a- Choose true or false: (1 mark for each)

- 1- You cannot use 1 byte (high or low bits) of ESP register, you can use either 16 or 32 bits (true)
2- The segment registers include SS, CS, DS, ES, DI, and SI (false)
3- Default segment cannot be overridden at all (false)
4- The statement MOV AX, [BP - 2] is incorrect (false)
5- MOV SS, CS is an incorrect instruction (true)

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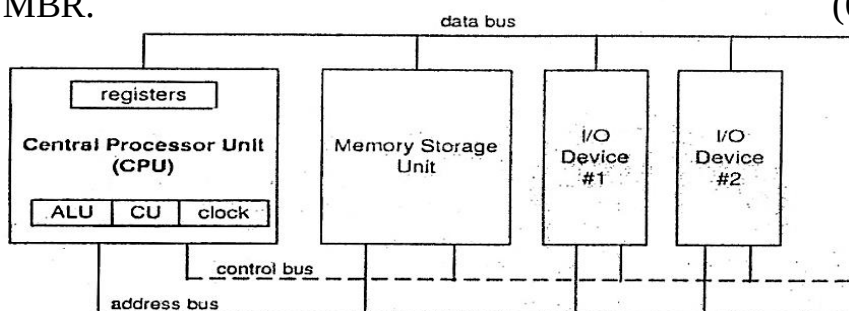
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Name:.....

Class:.....

Answer the following questions:

- 1- Draw a block diagram of a microcomputer, and then describe the functions of the registers PC and MBR. (6 Marks)

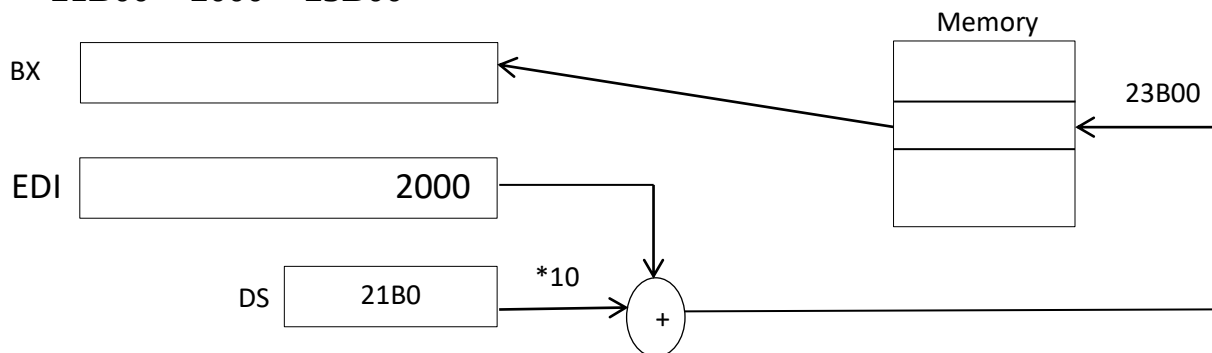


PC (Program Counter): Contains the address of an instruction to be fetched.

MBR (Memory Buffer Register): Contains a word of data to be written to memory or the word most recently read.

- 2- Given that SS = 10D0H, DS = 21B0H, DI = 2000H and BP = 0100H. Calculate and show by figure the memory address accessed by the following instruction: MOV BX, [DI] in direct addressing environment. (6 marks)

$$\begin{aligned} \text{Address} &= (\text{DS} \times 10) + \text{DI} \\ &= 21\text{B}00 + 2000 = 23\text{B}00 \end{aligned}$$



1- A debugger is

- a) easy to run and good with small programs b) difficult to run but good with small programs c) easy to run and good with long programs d) difficult to run and good with long programs

2- General purpose registers are

- a) 16 bit registers such as CS, DS b) 32 bit registers such as EAX, EBX c) 32 bit registers such as EFLAGS d) None

3- The operands in MOV instruction, source and destination in an instruction cannot be

- a) DS, ES b) AX, CS c) EAX, ESP d) BX, DX

4- The segment register of the instruction MOV 30[BP], BX will be:

- a) ES b) CS c) SS d) DS which contains 30

5- EPROM stands for:

- a) Electrically programmable read only memory b) Erasable programmable read write memory c) Erasable portable read only memory d) None of the mentioned

b- Choose true or false: (1 mark for each)

- 1- The statement MOV AX, [BP - 2] is correct (true)
2- The user visible registers include SS, CS, DS, ES, DI, and SI only (false)
3- A default segment can be overridden by writing the required segment before the offset (true)
4- You can use 1 byte (high or low bits) of ESI register by using SH, SL (false)
5- MOV [AX], [150H] is a correct instruction (false)